

Name: Maniram Chaudhary

Roll No: 28

Q. No. 1 (a) Evaluate the integral $\int_0^2 \int_{y^2}^4 y \cos(x)^2 dx dy$

Sol:-

The region 'R' is bounded by the curves.

$$x = y^2, \quad x = 4 \quad \& \quad y = 0, \quad y = 2$$

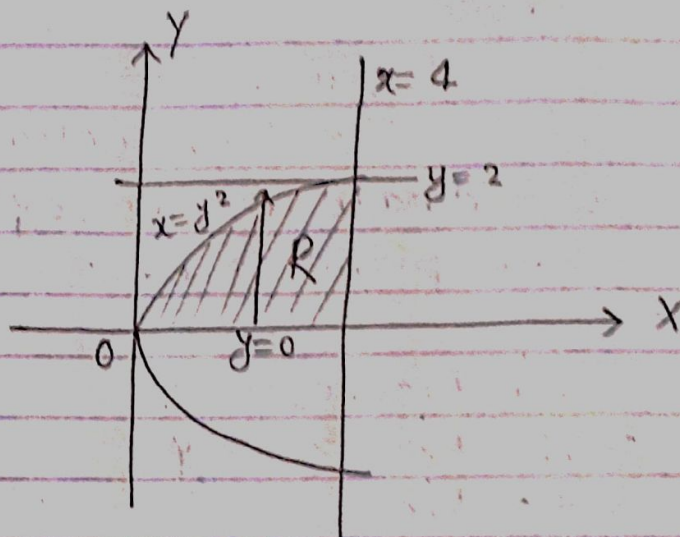
$$\therefore I = \int_0^4 \int_0^{\sqrt{x}} y \cos x^2 dy dx$$

$$= \int_0^4 \left[\cos x^2 \int_0^{\sqrt{x}} y dy \right] dx$$

$$= \int_0^4 \cos x^2 \left[\frac{y^2}{2} \right]_0^{\sqrt{x}} dx$$

$$= \int_0^4 \cos x^2 \cdot \frac{1}{2} [x - 0] dx$$

$$= \frac{1}{2} \int_0^4 x \cos x^2 dx$$



Put, $x^2 = t$

$$2x \frac{dt}{dx} = dt \Rightarrow 2x dx = dt \Rightarrow x dx = \frac{dt}{2}$$

When,

$$x = 0, \quad t = 0 \quad \& \quad x = 4, \quad t = 16$$

$$\therefore I = \frac{1}{4} \int_0^{16} \cos t dt$$

$$= \frac{1}{4} [\sin t]_0^{16}$$

$$= \frac{\sin 16}{4} //$$

Q.1 (b)

Find the volume in the first octant bounded by the co-ordinate planes, the cylinder $x^2 + y^2 = 9$ & the plane $z + y = 3$.

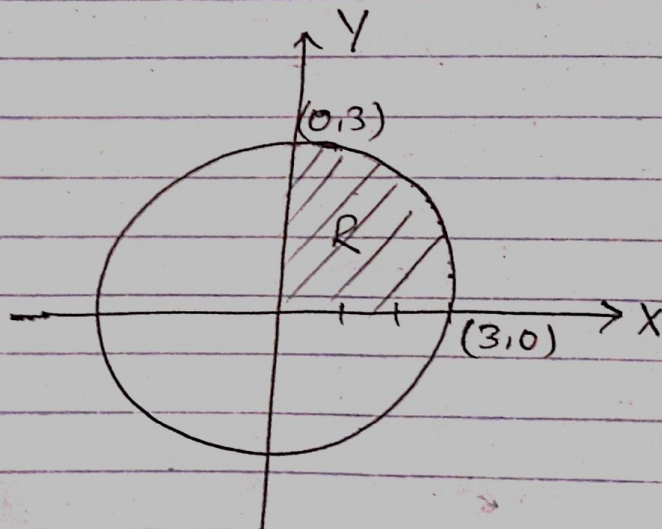
Soln:-

Given,

We have.

The first octant bounded by the coordinate planes, cylinder $x^2 + y^2 = 9$ and the plane $z + y = 3$.

From figure, it is clear that $z = 3 - y$ is to be integrated over the first quadrant of the circle $x^2 + y^2 = 9$.



For the base, limits for 'y' are $y = 0$ and $y = \sqrt{9 - x^2}$

limits for 'x' are $x = 0$ and $x = 3$

Now,

volume of the solid is

$$V = \int_0^3 \int_0^{\sqrt{9-x^2}} z \, dy \, dx$$

$$= \int_0^3 \int_0^{\sqrt{9-x^2}} (3-y) \, dy \, dx$$

$$= \int_0^3 \left[\int_0^{\sqrt{9-x^2}} (3-y) \, dy \right] dx$$

$$= \int_0^3 \left[3y - \frac{y^2}{2} \right]_0^{\sqrt{9-x^2}} dx$$

$$= \int_0^3 \left(3\sqrt{9-x^2} - \frac{9-x^2}{2} - (0-0) \right) dx$$

$$= \int_0^3 \left(3\sqrt{9-x^2} - \frac{9-x^2}{2} \right) dx$$

$$= \left[3 \left(\frac{x}{2} \sqrt{9-x^2} + \frac{9}{2} \sin^{-1} \left(\frac{x}{3} \right) \right) - \frac{1}{2} \left(9x - \frac{x^3}{3} \right) \right]_0^3$$

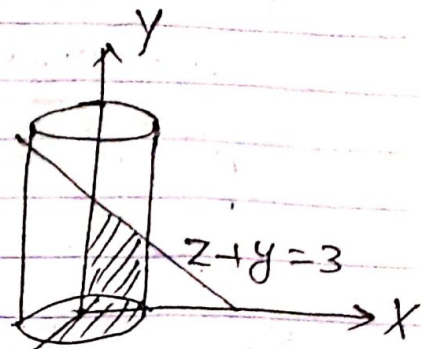
$$= \left[3 \left\{ \frac{3}{2} \sqrt{9-9} + \frac{9}{2} \sin^{-1} \left(\frac{3}{3} \right) \right\} - \frac{9 \times 3}{2} + \frac{27}{3} \right] - 3 \{ 0 + 0 - 0 \}$$

$$= 3 \left\{ \frac{9}{2} \cdot \frac{\pi}{2} - \frac{9}{2} \right\}$$

$$\left[\because \sin^{-1}(1) = \pi/2 \right]$$

$$= \frac{3}{2} \left[\frac{9\pi}{2} - 9 \right]$$

$$= \frac{27}{2} \left[\frac{\pi}{2} - 1 \right] //$$



Q.2 (a) Change the following integral into polar form and evaluate:

$$\int_{-1}^1 \int_0^{\sqrt{1-x^2}} \frac{2dy dx}{(1+x^2+y^2)^2}$$

Sol:-

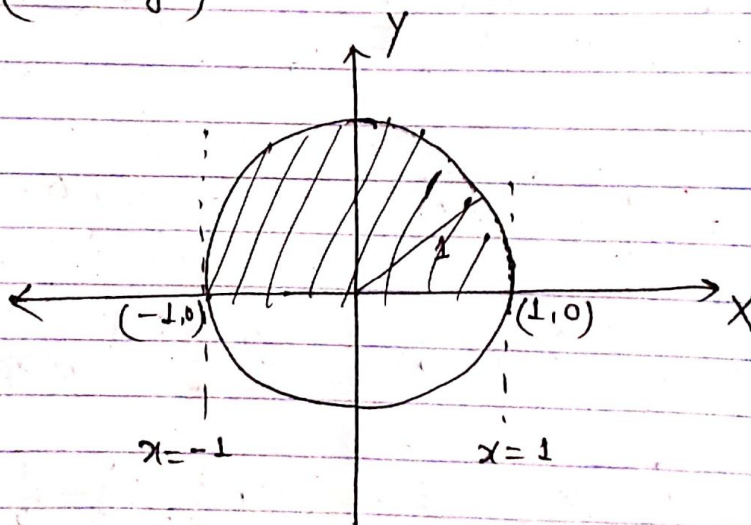
Given
Integral

$$I = \int_{-1}^1 \int_0^{\sqrt{1-x^2}} \frac{2dy dx}{(1+x^2+y^2)^2}$$

Here,
 $-1 \leq x \leq 1$ and
 $0 \leq y \leq \sqrt{1-x^2}$

Now,
changing the
region to polar,

$$x = r \cos \theta$$
$$\& y = r \sin \theta$$



Radius of circle is 1. So, $0 \leq r \leq 1$ and $\theta = 0$ to $\theta = \pi$
i.e. $0 \leq \theta \leq \pi$

Here,

$$(1+x^2+y^2)^2 = (1+r^2(\sin^2\theta + \cos^2\theta))^2$$
$$= (1+r^2)^2$$
$$r \neq 0$$
$$= (1+1)^2$$
$$= (2)^2 = 4$$

$$\therefore I = \int_0^\pi \int_0^1 \frac{2rdrd\theta}{4}$$

$$= \frac{1}{2} \int_0^\pi \left[\int_0^1 rdr \right] d\theta$$

$$= \frac{1}{2} \int_0^\pi \left[\frac{r^2}{2} \right]_0^1 d\theta$$

$$= \frac{1}{4} \int_0^\pi 1 d\theta$$

$$= \frac{1}{4} \left[\theta \right]_0^\pi$$

$$= \frac{\pi}{4} \underline{\underline{Ans}}$$

Q.2(b) Using power series method, solve $y'' + xy = 0$

Sol?.

$$y'' + xy = 0 \dots (i)$$

Let,

$$y = a_0 + a_1x + a_2x^2 + a_3x^3 + a_4x^4 + a_5x^5 + a_6x^6 + \dots \quad (ii)$$

the solution of eqⁿ (i)

Then,

$$y' = a_1 + 2a_2x + 3a_3x^2 + 4a_4x^3 + 5a_5x^4 + 6a_6x^5 + \dots$$

$$y'' = 2a_2 + 6a_3x + 12a_4x^2 + 20a_5x^3 + 30a_6x^4 + \dots$$

Substituting these values in eqⁿ (i)

$$y'' + xy = 0$$

$$\text{or, } (2a_2 + 6a_3x + 12a_4x^2 + 20a_5x^3 + 30a_6x^4 + \dots) + x(a_0 + a_1x + a_2x^2 + a_3x^3 + a_4x^4 + a_5x^5 + a_6x^6 + \dots)$$

$$\text{or, } [2a_2 + 6a_3x + 12a_4x^2 + 20a_5x^3 + 30a_6x^4 + \dots] + [a_0x + a_1x^2 + a_2x^3 + a_3x^4 + a_4x^5 + a_5x^6 + \dots]$$

$$\text{or, } 2a_2 + (6a_3 + a_0)x + (12a_4 + a_1)x^2 + (20a_5 + a_2)x^3 + (30a_6 + a_3)x^4 + \dots$$

Comparing on both side

$$2a_2 = 0$$

$$\Rightarrow a_2 = 0$$

$$6a_3 + a_0 = 0$$

$$a_0 = -6a_3$$

$$\Rightarrow a_3 = -\frac{a_0}{6}$$

$$\Rightarrow a_3 = -\frac{a_0}{3 \times 2}$$

$$12a_4 + a_1 = 0$$

$$a_1 = -12a_4$$

$$a_4 = -\frac{a_1}{12}$$

$$\Rightarrow a_4 = -\frac{a_1}{4 \times 3}$$

$$20a_5 + a_2 = 0$$

$$20a_5 = 0$$

$$a_5 = 0$$

$$30a_6 + a_3 = 0$$

$$30a_6 + \frac{a_0}{6} = 0$$

$$a_6 = -\left(\frac{a_0}{6 \times 30}\right)$$

Substituting these values in eqⁿ (ii)

$$y = a_0 + a_1x + a_2x^2 + a_3x^3 + a_4x^4 + a_5x^5 + a_6x^6 + \dots$$

$$= a_0 + a_1x + 0 \cdot x^2 + \left(-\frac{a_0}{6}\right)x^3 + \left(-\frac{a_1}{12}\right)x^4 + 0 \cdot x^5 + \left(+\frac{a_0}{180}\right)x^6 + \dots$$

$$y = a_0 + a_1 x - \frac{x^3 a_0}{3 \cdot 2} - \frac{x^4 a_1}{4 \times 3} + 0 + \frac{x^6 a_0}{6 \times 3 \times 0} + \dots$$

$$\therefore y = a_0 \left(1 - \frac{x^3}{6} + \frac{x^6}{180} + \dots \right) + a_1 \left(1 - \frac{x^4}{12} + \dots \right)$$

Ans

Q.No.3 a)

Write Legendre's differential equation and solve it by power series method.

Soln

Legendre's differential equation

$$(1-x^2)y'' - 2xy' + n(n+1)y = 0 \dots (i)$$

Let,

$$y = \sum_{m=0}^{\infty} a_m x^m \dots (ii) \quad \text{be the solution of eq}^n (i)$$

$$y' = \sum_{m=0}^{\infty} a_m \cdot m x^{m-1}$$

$$y'' = \sum_{m=0}^{\infty} a_m \cdot m(m-1) x^{m-2}$$

∴ From (i)

$$(1-x)^2 \sum_{m=0}^{\infty} a_m \cdot m(m-1) x^{m-2} - 2x \sum_{m=0}^{\infty} a_m \cdot m x^{m-1} - k \cdot \sum_{m=0}^{\infty} a_m x^m = 0$$

$$[\because k = n(n+1)]$$

$$\therefore \sum_{m=0}^{\infty} a_m \cdot m(m-1) x^{m-2} - \sum_{m=0}^{\infty} a_m \cdot m(m-1) x^m - 2 \sum_{m=0}^{\infty} a_m \cdot m x^m + k \sum_{m=0}^{\infty} a_m x^m = 0$$

$$a_m \cdot x^m = 0$$

$$\therefore \sum_{m=0}^{\infty} a_m \cdot m(m-1) x^{m-2} - \sum_{m=0}^{\infty} [m(m-1) + 2m - k] a_m x^m = 0$$

$$\therefore \sum_{m=0}^{\infty} a_m \cdot m(m-1) x^{m-2} - \sum_{m=0}^{\infty} (m^2 + m - k) a_m x^m = 0$$

Put, $m = s+2$ in first summation.

& $m = s$ in second summation.

We get

$$a_{s+2} (s+2)(s+2-1) x^{(s+2-2)} - (s^2 + s - k) a_s x^s = 0$$

$$\therefore a_{s+2} (s+2)(s+1) x^s - (s^2 - 3s + k) a_s x^s = 0$$

Equating the coefficient of x^s to zero.

$$a_{s+2} (s+2)(s+1) - \{s^2 + s - n(n+1)\} a_s = 0$$

$$a_{s+2} = \frac{(s^2 - s - n^2 - n) a_s}{(s+2)(s+1)}$$

$$= \frac{(s-n)(s+n) + (s-n) a_s}{(s+2)(s+1)}$$

$$= \frac{(s-n)(s+n+1) a_s}{(s+2)(s+1)}$$

$$\therefore a_{s+2} = - \frac{(n-s)(n+s+1) a_s}{(s+2)(s+1)}$$

When, $s = 0, 1, 2, 3, \dots$

$$s = 0$$

$$a_2 = - \frac{(n-0)(n+0+1) a_0}{(0+2)(0+1)} = - \frac{n(n+1) a_0}{2} = - \frac{n(n+1) a_0}{2!}$$

$$s = 1$$

$$a_3 = - \frac{(n-1)(n+1+1) a_1}{(1+2)(1+1)} = - \frac{(n-1)(n+2) a_1}{3 \cdot 2} = - \frac{(n-1)(n+2) a_1}{3!}$$

$$s = 2$$

$$a_4 = - \frac{(n-2)(n+2+1) a_2}{(2+2)(2+1)} = - \frac{(n-2)(n+3) a_2}{4 \cdot 3} = - \frac{(n-2)(n+3)}{4 \cdot 3} \left[- \frac{n(n+1) a_0}{2!} \right]$$

$$\therefore a_4 = \frac{n(n-2)(n+1)(n+3) a_0}{4!}$$

$$s = 3$$

$$a_5 = - \frac{(n-3)(n+3+1) a_3}{(3+2)(3+1)} = - \frac{(n-3)(n+4)}{5 \cdot 4} \left[- \frac{(n-1)(n+2) a_1}{3!} \right]$$

$$\therefore a_5 = \frac{(n-1)(n+2)(n-3)(n+4) a_1}{5!}$$

From (i)

$$y = \sum_{m=0}^{\infty} a_m x^m$$

$$= a_0 + a_1 x + a_2 x^2 + a_3 x^3 + a_4 x^4 + a_5 x^5 + \dots$$

$$= a_0 + a_1 x + \left[\frac{-n(n+1)a_0}{2!} \right] x^2 + \left[\frac{-(n-1)(n+2)a_1}{3!} \right] x^3 + \left[\frac{(n-2)(n+1)(n+3)a_0}{4!} \right] x^4 \\ + \left[\frac{(n-3)(n+1)(n+2)(n+4)a_1}{5!} \right] x^5 + \dots$$

$$= a_0 + a_1 x - \frac{n(n+1)a_0}{2!} x^2 - \frac{(n-1)(n+2)a_1}{3!} x^3 + \frac{(n+1)(n-2)(n+3)a_0}{4!} x^4 \\ + \frac{(n-1)(n+2)(n-3)(n+4)a_1}{5!} x^5 + \dots$$

$$= a_0 \left[1 - \frac{n(n+1)}{2!} x^2 + \frac{(n-2) \cdot n(n+1)(n+3)}{4!} x^4 - \dots \right] + a_1 \left[x - \frac{(n-1)(n+2)}{3!} x^3 + \frac{(n-1)(n+2)(n-3)(n+4)}{5!} x^5 \right] + \dots$$

$$y = a_0 y_1 + a_1 y_2$$

//

Q.3(b) If $J_n(x)$ is a Bessel's function, prove that
Soln.

$$(i) J_{-n}(x) = (-1)^n J_n(x)$$

We have,

$$J_n(x) = \sum_{m=0}^{\infty} \frac{(-1)^m x^{2m+n}}{2^{2m+n} \cdot m! (m+n)!}$$

$$\therefore J_{-n}(x) = \sum_{m=0}^{\infty} \frac{(-1)^m x^{2m-n}}{2^{2m-n} \cdot m! (m-n)!}$$

$$= \sum_{m=0}^{\infty} \frac{(-1)^m x^{2m-n}}{2^{2m-n} m! (m-n)!} \quad \left[\because \text{for } m < n \quad (m-n)! = \infty \right]$$

Put, $m-n=s$

$$\therefore m = s+n$$

$$\therefore J_{-n}(x) = \sum_{s=0}^{\infty} \frac{(-1)^{s+n} x^{2s+n}}{2^{2s+n} (s+n)! s!}$$

$$= (-1)^n \sum_{s=0}^{\infty} \frac{(-1)^s x^{2s+n}}{2^{2s+n} (s+n)! s!}$$

$$= (-1)^n \sum_{s=0}^{\infty} \frac{(-1)^s x^{2s+n}}{2^{2s+n} s! (s+n)!}$$

$$= (-1)^n J_n(x)$$

proved

$$(ii) \frac{d}{dx} [x^n J_n(x)] = x^n J_{n-1}(x)$$

Soln.

LHS

$$\frac{d}{dx} [x^n J_n(x)] = \frac{d}{dx} \left[\sum_{m=0}^{\infty} \frac{(-1)^m x^{2m+2n}}{2^{2m+n} m! (m+n)!} \right]$$

$$= \sum_{m=0}^{\infty} \frac{(-1)^m 2(m+n) \cdot x^{2m+2n-1}}{2^{2m+n} m! (m+n)!}$$

$$= \sum_{m=0}^{\infty} \frac{(-1)^m \cdot 2(m+n) \cdot x^{2m+2n-1}}{2^{2m+n} m! (m+n)(m+n-1)!}$$

$$= \sum_{m=0}^{\infty} \frac{(-1)^m \cdot x^{2m+2n-1}}{2^{2m+n-1} m! (m+n-1)!}$$

$$= \sum_{m=0}^{\infty} \frac{(-1)^m x^{2m+n-1} \cdot x^n}{2^{2m+n-1} m! (m+n-1)!}$$

$$= x^n J_{n-1}(x)$$

= RHS

proved.

Q. No. 4

a) Define Laplace transform. Find the Laplace transform of the following functions.

(i) $t^2 \sin \omega t$ (ii) $\frac{e^{-t} \cos t}{t}$ (iii) $t e^{-t} \cosh t$

Solⁿ:-

Laplace Transform:

Let $f(t)$ be a function of time (t) . Then Laplace transform of $f(t)$ is denoted by

$L\{f(t)\} = F(s)$ and given by

$$L\{f(t)\} = F(s) = \int_0^{\infty} e^{-st} f(t) dt$$

(i)

Solⁿ:-

Given,

$$t^2 \sin \omega t$$

Here,

$$L(\sin \omega t) = \frac{\omega}{s^2 + \omega^2}$$

$$\therefore L\{t^2 \sin \omega t\} = (-1)^2 \cdot \frac{d^2}{ds^2} \left[\frac{\omega}{s^2 + \omega^2} \right]$$

$$= \frac{d}{ds} \left[\frac{d}{ds} \left\{ \frac{\omega}{s^2 + \omega^2} \right\} \right]$$

$$= \frac{d}{ds} \omega \cdot \frac{d}{ds} \left(\frac{1}{s^2 + \omega^2} \right)$$

$$= -\frac{d}{ds} \omega [(s^2 + \omega^2)^{-2} 2s]$$

$$= -\frac{d}{ds} \frac{2\omega s}{(s^2 + \omega^2)^2}$$

$$= -2\omega \left[\frac{(s^2 + \omega^2)^2 - 2s(s^2 + \omega^2)2s}{(s^2 + \omega^2)^4} \right]$$

$$= -2\omega (s^2 + \omega^2) \left[\frac{s^2 + \omega^2 - 4s^2}{(s^2 + \omega^2)^4} \right]$$

$$= -2\omega (s^2 + \omega^2) \left[\frac{\omega^2 - 3s^2}{(s^2 + \omega^2)^3} \right]$$

$$\therefore L\{t^2 \sin \omega t\} = 2\omega \left[\frac{3s^2 - \omega^2}{(s^2 + \omega^2)^3} \right] //$$

$$(ii) \frac{e^{-t} \cos t}{t}$$

Soln.

$$L\left\{\frac{e^{-t} \cos t}{t}\right\} = \left[L\left\{\frac{\cos t}{t}\right\} \right]_{s \rightarrow s+1}$$

$$= \left[\int_s^\infty L\{\cos t\} ds \right]_{s \rightarrow s+1}$$

$$= \left[\int_s^\infty \frac{s}{s^2 + a^2} ds \right]_{s \rightarrow s+1}$$

$$= \left[\left[\frac{1}{2} \ln(s^2 + a^2) \right]_s^\infty \right]_{s \rightarrow s+1}$$

$$= \left[0 - \frac{1}{2} \ln(s^2 + a^2) \right]_{s \rightarrow s+1}$$

$$= -\frac{1}{2} \ln[(s+1)^2 + a^2]$$

$$\therefore L \left\{ \frac{e^{-t} \cos t}{t} \right\} = -\frac{1}{2} \ln[(s+1)^2 + a^2] //$$

(iii) $t e^{-t} \cosh t$

Sol?

$$\begin{aligned} L \{ t e^{-t} \cosh t \} &= \left[L \{ t \cosh t \} \right]_{s \rightarrow s+1} \\ &= \left[-\frac{d}{ds} L \{ \cosh t \} \right]_{s \rightarrow s+1} \\ &= - \left[\frac{d}{ds} \left(\frac{s}{s^2 - a^2} \right) \right]_{s \rightarrow s+1} \\ &= - \left[\frac{(s^2 - a^2) - s \cdot 2s}{(s^2 - a^2)^2} \right]_{s \rightarrow s+1} \\ &= - \left[\frac{s^2 - a^2 - 2s^2}{(s^2 - a^2)^2} \right]_{s \rightarrow s+1} \\ &= - \left[\frac{-a^2 - s^2}{(s^2 - a^2)^2} \right]_{s \rightarrow s+1} \\ &= \left[\frac{s^2 + a^2}{(s^2 - a^2)^2} \right]_{s \rightarrow s+1} \\ &= \left[\frac{1}{s^2 + a^2} \right]_{s \rightarrow s+1} \end{aligned}$$

$$\therefore L \{ t e^{-t} \cosh t \} = \frac{1}{(s+1)^2 + a^2} //$$

Q.No. 4 (b)

Find the inverse Laplace transform of the following functions:

(i) $\log\left(1 + \frac{\omega^2}{s^2}\right)$

Sol:-

Here,

$$\frac{d}{ds} F(s) = \frac{d}{ds} \log\left(\frac{s^2 + \omega^2}{s^2}\right)$$

$$= \frac{d}{ds} [\log(s^2 + \omega^2) - \log s^2]$$

$$= \frac{1}{s^2 + \omega^2} \cdot 2s - \frac{1}{s^2} \cdot 2s$$

$$= \frac{2s}{s^2 + \omega^2} - \frac{2s}{s^2}$$

$$= 2 \left[\frac{s}{s^2 + \omega^2} \right] - 2 \cdot \frac{1}{s}$$

$$= 2 L\{\cos \omega t\} - 2 L\{1\}$$

$$= L\{2 \cos \omega t - 2\}$$

$$\therefore L\{t \cdot f(t)\} = -\frac{d}{ds} F(s)$$

$$= -[L\{2 \cos \omega t - 2\}]$$

$$L\{t \cdot f(t)\} = L\{2 - 2 \cos \omega t\}$$

$$\{t \cdot f(t)\} = 2 - 2 \cos \omega t$$

$$f(t) = \frac{2 - 2 \cos \omega t}{t} //$$

$$(ii) \quad \frac{1}{s(s^2 + \omega^2)}$$

Sol:-

$$f(t) = \left\{ \frac{1}{s(s^2 + \omega^2)} \right\}$$

We know,

$$L^{-1} \left\{ \frac{1}{s} \right\} = 1 \quad \& \quad L^{-1} \left\{ \frac{1}{s^2 + \omega^2} \right\} = \frac{1}{\omega} \sin \omega t$$

Now, By convolution theorem

$$L^{-1} \left\{ \frac{1}{s(s^2 + \omega^2)} \right\} = \int_0^t 1 \cdot \frac{\sin \omega(t-u)}{\omega} du$$

$$= \frac{1}{\omega} \int_0^t \sin \omega(t-u) du$$

$$= \frac{1}{\omega} \int_0^t \sin(\omega t - \omega u) du$$

$$= \frac{1}{\omega} \left[\frac{-\cos(\omega t - \omega u)}{-\omega} \right]_0^t$$

$$= \frac{1}{\omega^2} [\cos(\omega t - \omega t) - \cos \omega t]$$

$$= \frac{1}{\omega^2} [\cos 0 - \cos \omega t]$$

$$= \frac{1}{\omega^2} (1 - \cos \omega t)$$

$$\therefore L^{-1} \left\{ \frac{1}{s} \cdot \frac{1}{(s^2 + \omega^2)} \right\} = \frac{1}{\omega^2} (1 - \cos \omega t) \quad //$$

Q. No 5 (a)

Solve the initial value problem using Laplace Transform:

$$y'' + 2y' - 3y = 6e^{-2t}, \quad y(0) = 2, \quad y'(0) = -14.$$

Sol:-

Given,

$$y'' + 2y' - 3y = 6e^{-2t} \quad \dots (i)$$

Taking Laplace transform on both side

$$L\{y'' + 2y' - 3y\} = L\{6e^{-2t}\}$$

$$\therefore L\{y''\} + 2L\{y'\} - 3L\{y\} = 6L\{e^{-2t}\}$$

$$\therefore [s^2 L(y) - sy(0) - y'(0)] + 2[sL(y) - y(0)] - 3L(y) = 6 \cdot \frac{1}{s+2}$$

$$\therefore L\{y\} [s^2 + 2s - 3] - 2s + 14 - 4 = \frac{6}{s+2}$$

$$\therefore L\{y\} [s^2 + 2s - 3] - 2s + 10 = \frac{6}{s+2}$$

$$\therefore L\{y\} [s^2 + 2s - 3] - 2s + 10 = \frac{6}{s+2}$$

$$\therefore L\{y\} [s^2 + 2s - 3] = \frac{6}{s+2} + (2s - 10)$$

$$\therefore L\{y\} = \frac{6 + 2s(s+2) - 10(s+2)}{(s+2)(s^2 + 2s - 3)}$$

$$\therefore L\{y\} = \frac{6 + 2s^2 + 4s - 10s - 20}{(s+2)(s^2 + 2s - 3)}$$

$$\begin{aligned} \therefore L\{y\} &= \frac{2(s^2 - 3s - 14)}{(s+2)(s^2 + 2s - 3)} \\ &= \frac{2(s^2 - 3s - 14)}{(s+2)(s+3)(s-1)} \end{aligned}$$

Now,

$$\frac{s^2 - 3s - 14}{(s+2)(s+3)(s-1)} = \frac{A}{s+2} + \frac{B}{s+3} + \frac{C}{s-1} \quad \dots (ii)$$

$$\begin{aligned} s^2 - 3s - 14 &= A(s+3)(s-1) + B(s+2)(s-1) + C(s+2)(s-1) \\ &= A(s^2 + 2s - 3) + B(s^2 + s - 2) + C(s^2 - s - 6) \end{aligned}$$

$$\therefore s^2 - 3s - 14 = (A+B+C)s^2 + (2A+B-C)s + (-3A-2B-6C)$$

Comparing on both sides

$$A+B+C=1, \quad 2A+B-C=-3 \quad -3A-2B-6C=-14$$

On solving,

$$\text{We get, } A = \frac{4}{3}, \quad B = -3 \quad \text{and } C = \frac{8}{3}$$

Then,

$$\begin{aligned} L\{y\} &= \frac{4}{3} \cdot \frac{1}{s+2} + \left(\frac{-3}{s+3} \right) + \frac{8}{3} \left(\frac{1}{s-1} \right) \\ &= \frac{4}{3} L\{e^{-2t}\} - 3L\{e^{-3t}\} + \frac{8}{3} L\{e^t\} \\ &= L\left[\frac{4}{3} e^{-2t} - 3e^{-3t} + \frac{8}{3} e^t \right] // \end{aligned}$$

Q. No 5 (b) state & prove first shifting theorem of Laplace transform. Use it to evaluate the Laplace transform of $e^{-2t} \sin nt$.

Sol:-

First shifting theorem:

Statement:

If the Laplace transform of $f(t)$ is $F(s)$ then Laplace transform of $[e^{at} f(t)]$ is $F(s-a)$, where,

$$s-a > k$$

$$s > k$$

$$\text{i.e. } L\{e^{at} f(t)\} = [F(s)]_{s \rightarrow s-a}$$

$$\equiv F(s-a)$$

Proof:

By the definition of Laplace Transform

$$L\{f(t)\} = \int_0^{\infty} e^{-st} \cdot f(t) dt \quad \dots (i)$$

$$\therefore L\{e^{at} \cdot f(t)\} = \int_0^{\infty} e^{-st} \cdot e^{at} \cdot f(t) dt$$

$$= \int_0^{\infty} e^{-(s-a)t} \cdot f(t) dt$$

$$= \int_0^{\infty} e^{-(s-a)t} f(t) dt \quad \dots (ii)$$

$$\therefore L\{e^{at} f(t)\} = F(s-a)$$

||y||
 $\therefore L\{e^{-at} \cdot f(t)\} = F(s+a) = [F(s)]_{s \rightarrow s+a}$ proved

Given,
 $L\{e^{-2t} \sin \pi t\}$

We know,

$$L\{\sin \pi t\} = \frac{\pi}{s^2 + \pi^2}$$

$$\therefore L\{e^{-2t} \sin \pi t\} = \left[\frac{\pi}{s^2 + \pi^2} \right]_{s \rightarrow s+2}$$

$$= \frac{\pi}{(s+2)^2 + \pi^2}$$

$$= \frac{\pi}{s^2 + 4s + 4 + \pi^2}$$

$$= \frac{\pi}{s^2 + 4s + 4 + \pi^2} //$$

Q. No. 6

a) Define Fourier series of a function of period 2π .
Find the Fourier series of

$$f(x) = \begin{cases} 1 & \text{for } -\pi < x < 0 \\ -1 & \text{for } 0 < x < \pi \end{cases}$$

Sol:-

Let, $f(x)$ be a function defined in the interval $(\alpha, \alpha + 2\pi)$. Then, Fourier series of $f(x)$ is

$$f(x) = a_0 + \sum_{n=1}^{\infty} (a_n \cos nx + b_n \sin nx) \quad \dots (i)$$

Where, a_0 , a_n & b_n are called Euler's constant & given by

$$a_0 = \frac{1}{2\pi} \int_{\alpha}^{\alpha+2\pi} f(x) dx$$

$$a_n = \frac{1}{\pi} \int_{\alpha}^{\alpha+2\pi} f(x) \cdot \cos nx dx$$

$$\& \quad b_n = \frac{1}{\pi} \int_{\alpha}^{\alpha+2\pi} f(x) \cdot \sin nx dx$$

Given,

$$f(x) = \begin{cases} 1 & \text{for } -\pi < x < 0 \\ -1 & \text{for } 0 < x < \pi \end{cases}$$

Here,

$$\begin{aligned} \text{Period} &= \pi - (-\pi) \\ &= 2\pi \end{aligned}$$

$\therefore$ Fourier series of $f(x)$ is

$$f(x) = a_0 + \sum_{n=1}^{\infty} [a_n \cos nx + b_n \sin nx]$$

So,

$$a_0 = \frac{1}{2\pi} \int_{-\pi}^{\pi} f(x) dx$$

$$= \frac{1}{2\pi} \left[\int_{-\pi}^0 1 dx + \int_0^{\pi} -1 dx \right]$$

$$= \frac{1}{2\pi} \left[(x)_{-\pi}^0 + (-x)_0^{\pi} \right]$$

$$= \frac{1}{2\pi} [\pi - \pi]$$

$$\therefore a_0 = 0$$

$$a_n = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \cos nx \, dx$$

$$= \frac{1}{\pi} \left[\int_{-\pi}^0 \cos nx \, dx + \int_0^{\pi} \cos nx \, dx \right]$$

$$= \frac{1}{\pi} \left[\left\{ \frac{\sin nx}{n} \right\}_{-\pi}^0 - \left\{ \frac{\sin nx}{n} \right\}_0^{\pi} \right]$$

$$= \frac{1}{\pi} \left[\frac{1}{n} [\sin 0 - \sin(-n\pi)] - \frac{1}{n} [\sin n\pi - \sin 0] \right]$$

$$= \frac{1}{\pi} \left[\frac{1}{n} [0 + \sin n\pi] - \frac{1}{n} [\sin n\pi - 0] \right]$$

$$= \frac{1}{n\pi} [\sin n\pi - \sin n\pi]$$

$$\therefore a_n = 0$$

$$b_n = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \sin nx \, dx$$

$$= \frac{1}{\pi} \left[\int_{-\pi}^0 \sin nx \, dx + \int_0^{\pi} -\sin nx \, dx \right]$$

$$= \frac{1}{\pi} \left\{ \left[-\frac{\cos nx}{n} \right]_{-\pi}^0 + \left[-\left(-\frac{\cos nx}{n} \right) \right]_0^{\pi} \right\}$$

$$= \frac{1}{n\pi} \left\{ -\cos 0 - (\cos(-n\pi)) + [\cos n\pi - \cos 0] \right\}$$

$$= \frac{1}{n\pi} [-1 + \cos n\pi + \cos n\pi - 1]$$

$$= \frac{1}{n\pi} [2 \cos n\pi - 2]$$

$$= \frac{2}{n\pi} [(-1)^n - 1]$$

$$\therefore b_n = \frac{2}{n\pi} [(-1)^n - 1]$$

$$\therefore f(x) = \sum_{n=1}^{\infty} \left\{ \frac{2}{n\pi} [(-1)^n - 1] \cdot \sin nx \right\} //$$

Q-6(b) Find the Fourier series of

$$f(x) = \begin{cases} k & \text{for } -\pi/2 < x < \pi/2 \\ 0 & \text{for } \pi/2 < x < 3\pi/2 \end{cases}$$

and show that $1 - \frac{1}{3} + \frac{1}{5} - \frac{1}{7} + \dots = \frac{\pi}{4}$

Sol:- Period = $\frac{3\pi}{2} - (-\frac{\pi}{2}) = 2\pi$

Fourier series of $f(x)$ is

$$f(x) = a_0 + \sum_{n=1}^{\infty} (a_n \cos nx + b_n \sin nx) \dots (i)$$

So,

$$a_0 = \frac{1}{2\pi} \left[\int_{-\pi/2}^{\pi/2} k \, dx + \int_{\pi/2}^{3\pi/2} 0 \, dx \right]$$

$$= \frac{1}{2\pi} k \left[x \right]_{-\pi/2}^{\pi/2} + 0$$

$$= \frac{k}{2\pi} \left[\frac{\pi}{2} - \left(-\frac{\pi}{2}\right) \right]$$

$$= \frac{k}{2\pi} \pi$$

$$\therefore a_0 = \frac{k}{2}$$

$$a_n = \frac{1}{\pi} \left[\int_{-\pi/2}^{\pi/2} k \cos nx \, dx \right] + 0$$

$$= \frac{k}{\pi} \left[\frac{\sin nx}{n} \right]_{-\pi/2}^{\pi/2}$$

$$= \frac{k}{n\pi} \left[\sin \frac{n\pi}{2} - \sin \left(-\frac{n\pi}{2}\right) \right]$$

$$= \frac{k}{n\pi} \left[\sin \frac{n\pi}{2} + \sin \frac{n\pi}{2} \right] = \frac{2k}{n\pi} \sin \left(\frac{n\pi}{2}\right)$$

~~$$a_n = \frac{k}{n\pi} \left[\sin \frac{n\pi}{2} + \sin \frac{n\pi}{2} \right] = \frac{2k}{n\pi} \sin \left(\frac{n\pi}{2}\right)$$~~

$$b_n = \frac{1}{\pi} \left[\int_{-\pi/2}^{\pi/2} k \sin nx \, dx \right] + 0$$

$$= \frac{k}{\pi} \left[\frac{-\cos nx}{n} \right]_{-\pi/2}^{\pi/2}$$

$$= \frac{-k}{n\pi} \left[\cos \frac{n\pi}{2} - \cos \frac{n\pi}{2} \right]$$

$$\therefore b_n = 0$$

From (i)

$$f(x) = \frac{k}{2} + \sum_{n=1}^{\infty} \left[\frac{2k}{n\pi} \sin \frac{n\pi}{2} \cosh nx \right]$$

$$= \frac{k}{2} + \frac{2k}{\pi} \cos x + \frac{2k}{3\pi} \cos 3x + \frac{2k}{5\pi} \cos 5x + \dots$$

~~$\frac{2k}{n\pi} \sin \frac{n\pi}{2}$~~

$$f(x) = \frac{k}{2} + \frac{2k}{\pi} \cos x - \frac{2k}{3\pi} \cos 3x + \frac{2k}{5\pi} \cos 5x - \dots \quad (ii)$$

$\therefore \cos 0 = 1$, so, taking $x=0$ in eqⁿ eqⁿ (ii)
We get,

$$f(0) = \frac{k}{2} + \frac{2k}{\pi} \left[\cos 0 - \frac{\cos 3 \cdot 0}{3} + \frac{\cos 5 \cdot 0}{5} - \dots \right]$$

$$k = \frac{k}{2} + \frac{2k}{\pi} \left[1 - \frac{1}{3} + \frac{1}{5} - \dots \right]$$

$$k - \frac{k}{2} = \frac{2k}{\pi} \left(1 - \frac{1}{3} + \frac{1}{5} - \dots \right)$$

$$\frac{k}{2} = \frac{2k}{\pi} \left(1 - \frac{1}{3} + \frac{1}{5} - \dots \right)$$

$$\therefore \frac{\pi}{4} = 1 - \frac{1}{3} + \frac{1}{5} - \frac{1}{7} + \dots$$

proved

Q.7 Attempt All

(i) Find the smallest period of the function $\sin \pi x$.

Sol?

Let,

$T > 0$ be the period of $f(x)$.

So that, $f(x+T) = f(x)$

$$\text{i.e. } \sin \pi(x+T) = \sin \pi x \cos \pi T + \cos \pi x \sin \pi T \\ = \sin \pi x$$

On comparing,

$$\cos \pi T = 1$$

$$\hookrightarrow \cos \pi T = \cos 2\pi$$

$$[\because \cos 2\pi = 1]$$

$$\hookrightarrow \pi T = 2\pi$$

$$\therefore T = 2$$

Period of $\sin \pi x = 2$ //

(ii) Evaluate: $\int_0^2 \int_0^4 (4-x^2) dx dy$

Sol?

$$\text{Let } I = \int_0^2 \int_0^4 (4-x^2) dx dy$$

$$= \int_0^2 \left[\int_0^4 (4-x^2) dx \right] dy$$

$$= \int_0^2 \left[4x - \frac{x^3}{3} \right]_0^4 dy$$

$$= \int_0^2 \left[4 \times 4 - \frac{64}{3} - (4 \times 0 - \frac{0}{3}) \right] dy$$

$$= \int_0^2 \left(16 - \frac{64}{3}\right) dy$$

$$= \int_0^2 \left(\frac{48-64}{3}\right) dy$$

$$= -\frac{16}{3} \left[\int_0^2 dy \right]$$

$$= -\frac{16}{3} [y]_0^2$$

$$= -\frac{16}{3} \times (2-0)$$

$$\therefore I = -\frac{32}{3} //$$

(iii) Draw the graph of the Legendre polynomials $P_2(x)$ & $P_3(x)$.

Sol?

Legendre's Polynomials

$$P_n(x) = \sum_{m=0}^{\infty} \frac{(-1)^m (2n-2m)!}{2^n m! (n-m)! (n-2m)!} x^{n-2m}$$

Now,

$$P_2(x) = \frac{4!}{2^2 (2!)^2} x^2 - \frac{2!}{2^2 (1!) (0!)} x^0$$

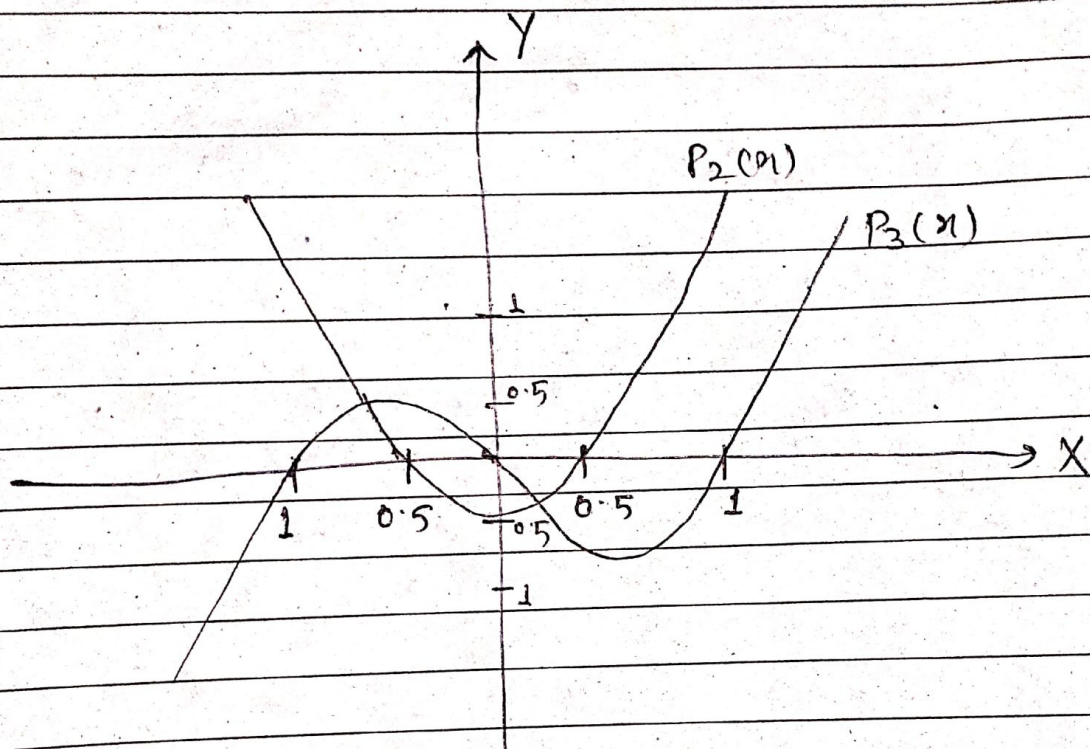
$$= \frac{3}{2} x^2 - \frac{1}{2}$$

$$P_2(x) = \frac{1}{2} (3x^2 - 1)$$

$$P_3(x) = \frac{6!}{2^3(3!)^2} x^3 - \frac{4!}{2^3(2!)^2} x$$

$$= \frac{5}{2} x^3 - \frac{3}{2} x$$

$$= \frac{1}{2} (5x^3 - 3x)$$



(v) Find the Laplace transform of unit step function.

Sol: Unit step function $f(x) = \begin{cases} 0 & \text{if } t < a \\ 1 & \text{if } t \geq a \end{cases}$

$$\text{Now, } L\{u_a(t)\} = \int_0^a e^{-st} \cdot u_a(t) dt$$

$$= \int_0^a e^{-st} \cdot 0 dt + \int_a^\infty e^{-st} \cdot 1 dt$$

$$= \left[\frac{e^{-st}}{-s} \right]_a^\infty = -\frac{1}{s} [e^{-\infty} - e^{-as}]$$

$\therefore L\{u_a(t)\} = \frac{e^{-as}}{s}$ is the Laplace transform of unit step function.